45. Operators are tested for their efficiency in terms of number of units produced per day by five different types of machines. Test at 5% level of significance whether the operators and machines differ in terms of their efficiency?

Operators	Machine types				
	A	B	C	D	E
I	8	10	7	12	6
II	12	13	8	9	12
III	7	8	6	8	8
IV	5	5	3	5	14

ANSWERS

I. 1. (c) 2. (b) 3. (d) 4. (a) 5. (a)
6. (c) 7. (a) 8. (c) 9. (a) 10. (a)
11. (c) 12. (c) 13. (b) 14. (c) 15. (d)
16. (a) 17. (c) 18. (b) 19. (c) 20. (c)

III. 36. $F_0 = 1.06$, H_0 is not rejected
37. $F_0 = 1.02$, H_0 is not rejected
38. $F_0 = 1.25$, H_0 is not rejected
39. $F_0 = 1.34$, H_0 is not rejected
40. $F_0 = 2.52$, H_0 is not rejected
41. $F_0 = 4.59$, H_0 is rejected
42. $F_0 = 1.097$, H_0 is not rejected
43. $F_0 = 3.33$, H_0 is not rejected
44. F_0 =2.39, H_0 is not rejected $F_0 = 3.59$, H_0 is not rejected
45. $F_0 = 2.53$, H_0 is not rejected $F_0 = 1.24$, H_0 is not rejected

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